RoBenDevs Product Manager Internship
Day 1 (Thursday: 5/11/2026)

Agile Methodology (Definition and Basics):

Agile Methodology is a product development lifecycle framework, predominantly used in software development, which emphasizes iterative development, incremental delivery, customer collaboration, continuous feedback, and responding to changing requirements.

Derived from the Agile Manifesto of 2001, it focuses on 5 C's as core principles, *Communication, Collaboration, Commitment, Customer and Continuous Improvement* to harbour a setting for open communication, engaging teamwork and continuous focused improvement to deliver the best product for customer satisfaction.

Typical Agile framework involves:

1. **Planning:** Defining goals and requirement based on customer needs, thus planning further work
2. **Development:** Collaborative development to produce selected functionalities
3. **Testing/Validation:** Testing the product via comparing with requirements declared before, finding defects and solving them ahead of delivering
4. **Incremental Delivery:** Delivering the product in increments, taking feedback from stakeholders in due process and improving the overall quality of both the product and process for maximum stakeholder satisfaction.
5. **Review and Feedback:** Evaluating the work and taking feedback from stakeholders and client to guide further development
6. **Adaptation and Improvement**: Adapting the process, requirements or priorities based on feedback from stakeholders.

Some of the most used Agile Frameworks are Scrum, Kanban, Extreme Programming(XP) which are mostly used in software development.

Scrum:

Scrum is a widely used agile management framework that helps teams organize and oversee their work through collaboration over iterative work. It emphasizes an iterative and incremental approach, thus helping teams to self organize and collaborate to deliver High quality software even if the product is complex.

Scrum follows the agile framework and Refines it, makes it more executable. Scrum uses sprints, which last from two to four weeks, depending on the requirements.

Scrum follows a 3-5-3 rule that defines the roles, framework and requirements (named as artifacts) of scrum. These are,

3 Roles:

1. **Product owner**: They are responsible for understanding the business, customer and market requirements. They communicate the product goal and communicate the product backlogs with the rest of the stakeholders. They represent the customer.
2. **Scrum Master:** Are responsible for coaching teams on self management and collaboration, product owners on effective goal declaration and communication, and the business by leading on the Scrum process and look for ways to fine-tune their practice.
3. **Development Team:** The team that designs and develops the product.

Scrum Teams are always to be a small, nimble yet tight-knit group roughly around 10 people.

5 Events:

1. **Sprint:** A short time boxed period that the scrum team uses to complete a set amount of work. It can be defined as a short project within the domain of a larger project. It must have clear goals, a defined body of work for proper planning,execution, review and later on improvements.
2. **Sprint Planning:** The first event of scrum framework and subsequently also a sprint, A sprint planning defines the work that needs to be done within a sprint. It connects product backlog to daily team work.
3. **Daily Scrum:** Also known as daily Standups or sitdown, it focuses on the development team conveying their progress, todos and blockers to the product owner or scrum master
4. **Sprint Review:** A collaborative working session held at the end of a sprint where the Scrum Team and key stakeholders inspect the product increment, demonstrate completed work, and adapt the Product Backlog
5. **Sprint Retrospective:** An event where the scrum team inspects how the last Sprint went with regards to individuals, interactions, processes, tools, and their Definition of Done.

3 Artifacts:

1. **Product Backlog:** Product Backlog is the primary dynamic list of features, requirements, enhancements, and fixes that the product owner or product manager needs to do and maintain. It dictates the sprint backlog.
2. **Sprint Backlog:** Sprint Backlog is the list of items, user stories, or bug fixes, selected by the development team from the product backlog for implementation in the current sprint cycle.
3. **Increment/Definition of Doneness:** Defines how you define your sprint goals as done, defers organization to organization, Example: some may say release of a sprint is increment others may say testing and review as a definition of doneness.

Sprints:

Sprints are short, time-boxed periods when a Scrum team works to complete a set amount of work. In effect, each sprint can be considered a short project. It has a clear goal, a defined body of work, and a complete cycle of planning, execution, review, and improvement.

Common Distinguish between Scrum and Sprits are that Scrums is a framework for managing complex work while sprint is a fixed length cycle where team plans, reviews and reflects on the work, working towards a sprint goal and delivering a usable product increment.

The Key aspects of Sprints are:

1. Fixed Duration

2. Clear Goals
3. Commitment towards a shippable work

Spring Structures:

There are a few events that define the structure of how a sprint is conducted. They are:

1. ***Sprint Planning:*** It's the first event in the Scrum framework and also in sprints. It happens before the sprint starts and here defines the work to be completed in a sprint.

   Here the team takes a takes task from the predefined and reorganized product backlog, and the development team along with product owner and other team members, select the tasks that needs to be done in the upcoming sprint

   The participants here are:
   a. **Product owner:** Defines goals based on seeked value
   b. **Development team:** Understand how they can or can't deliver that goal

   There are a few steps:
   I. Updating the backlog for better sprint preparation and better stakeholder involvement and feedback management
   II. Timeboxing no more than two hours for each weeks sprint planning
   III. Properly Defining User Stories and customer point of view
   IV. Get rid of as much unknowns as possible while estimating time for more accurate time prediction

2. **Daily standups:**
   The daily stand-up is a short, daily meeting to discuss progress and identify blockers. The product manager sits down with the corresponding Dev team members and all stakeholders answer these three questions. It's typically held every morning and takes no more than 15 minutes.

   What did I work on yesterday?
   What am I working on today?
   What issues are blocking me?

   Daily standups make sure any blockers are visible as soon as possible and they are resolved in daily manner. It also helps navigate the daily work.

3. **Sprint Review:**

   In a sprint review, the development team and stakeholders gather to review and demonstrate the work completed during the sprint.

   1. Key participants in a sprint review include the Scrum team (developers, Scrum Master, and product owner) and relevant stakeholders such as customers, business owners, or subject matter experts. The product owner typically leads the session, ensuring that feedback is captured and next steps are clear.

Main purpose of a sprint review:

1. **Demonstration of completed work:** Showing what was built during the sprint
2. **Feedback collection:** Gathering input from stakeholders
3. **Alignment with product vision:** Making sure the work still supports the product's direction
4. **Celebration of achievements:** Recognizing the team's accomplishments
5. **Identification of improvements:** Spotting what can be adjusted going forward

What occurs during the sprint review:

a. The Scrum team presents completed work to stakeholders, demonstrating new features or improvements and gathering feedback
b. The team discusses what was accomplished, what remains unfinished, and any changes to the product backlog
c. The stakeholders provide feedback, discuss the backlog updates, ask questions and suggest adjustments based on real progress
d. The agenda starts with a welcome and overview of the sprint goal, followed by demonstrations of completed work
e. It usually concludes with updates to the product backlog and a discussion of next steps

4. Sprint Retrospective

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly. While retrospectives are most often associated with software development teams, their value extends far beyond the software industry.

The key participants of a Retrospective are , Scrum team (developers, Scrum Master, and product owner) and relevant stakeholders such as customers, business owners, or subject matter experts. The product owner typically leads the session, ensuring that feedback is captured and next steps are clear.

The purpose of the retrospective meeting is to:

1. Evaluate how the last sprint, iteration, or work item went, specifically around the team dynamic, processes, and tools
2. Articulate and stack rank the items that went well, and those that did not
3. Create and implement a plan for improving the way the team does work

How to run your first retrospective:

a. **The when:** For agile teams working in the traditional two-week sprint, the retrospective should take place at the end of every sprint. For teams running a more Kanban-esque style of work, a monthly or quarterly retrospective may make more sense, as may one after major initiatives have been rolled out.

b. **The who:** Every team member should attend, with a facilitator leading the discussion. The facilitator can be the Scrum Master, the product owner, or the role can rotate throughout the team.
c. **The what:**
    i. Create a short list of things that worked well and things that could be improved
    ii. Prioritize this list by importance as a team.
    iii. Discuss ways and tactics to improve the top two items on the "room for improvement" list.
    iv. Create an action plan.
    v. Be disciplined about executing step 4.

Vary the list prompts. At the end of the day, the retrospective is meant to uncover what's working and what's not. Consider these different prompts:

1. ***Start / Stop / Continue:*** What the team should start doing, stop doing, and continue doing. Focus on ways to discontinue items in the "Stop" column.
2. ***More / Less:*** What the team needs to do more and less of. Create a plan around how to tackle the top items in the "do less" list.
3. ***Glad / Sad / Mad:*** What made the team glad, sad, and mad. Focus on the sad and mad lists and how to improve, so there are only items in the glad column next time.

Scrum Roles:

Scrum roles define the participant of a scrum. There are mainly three roles in scrum:

1. **Product Owner:**
   The product owner is the main person for the product. He is who understands the customer and business requirements, then creates and manages the product backlog based on those requirements. He should be dynamic and should not only understand the customer but also have a vision for the value the Scrum team is delivering to the customer.There should be only one product owner per product.

   The Product Owner generally leads all scrum meetings. They are responsible for portraying the clients perspective to the team so that the clients requirements are met every step of the process.

2. **Scrum Master:**
   Scrum Master is the role responsible for gluing everything together and ensuring that Scrum is being done well by acting as the facilitator and coach of the team, fostering collaboration and continuous improvement to help deliver value efficiently. They don’t directly influence the main product decisions but oversee the decision making process and thus help the teams.The are responsible for removing unnecessary obstacles that might impede progress, and thus shields the team from distractions
3. **Development Team:**
   The development team can be composed of all kinds of people, including designers, writers, programmers, etc who are responsible for delivering the work through the sprint. They are the people who are responsible for developing the product and implementing the sprint backlogs.

To ensure transparency during the sprint, they meet daily at the daily scrum. They also participate in sprint backlog meetings to determine sprint backlog and sprint length and also participate in sprint review and retrospective at the end of sprint.

Scrum Artifacts:

Scrum artifacts are information that the Scrum Team and stakeholders use to detail the product being developed, the actions required to produce it, and the work performed during the project.

Scrum defines three main artifacts. Which are

1. **Product Backlog:**
   The Product Backlog is an ordered and continuously updated list of everything that is needed to improve and develop a product. It includes new features, enhancements, bug fixes, technical tasks, and other work requirements.It's compiled from different input sources like customer support, competitor analysis, market demands, stakeholder feedback, and general business analysis.The Product Backlog is a "live" artifact in that it is updated whenever new information, requirements, or priorities arise.The Product Owner is accountable for maintaining and ordering the Product Backlog, ensuring that the most valuable and relevant work is prioritized.

   Commitment – Product Goal:
   The Product Goal defines the future state of the product that the Scrum Team is working towards. It provides a long-term objective and helps the team understand what they are trying to achieve through their development efforts.

2. **Sprint Backlog:**
   The Sprint Backlog is a set of Product Backlog items selected for implementation during the current sprint, along with a plan for delivering them. Sprint Backlogs are created by the Developers during Sprint Planning to determine what work can be completed within the sprint and how that work will be accomplished. The Sprint Backlog is not a fixed list. Developers can update it throughout the sprint as they learn more about the work and adjust their plans accordingly. Consider an example task like "build a shopping cart page", which requires multiple design and development subtasks. The Product Backlog may contain the main requirement, while the Sprint Backlog contains the selected work and detailed plan required to implement it.

   Commitment – Sprint Goal:
   The Sprint Goal defines the main objective that the team aims to achieve during a sprint. It provides direction and flexibility, allowing the Developers to adjust the exact tasks while remaining focused on the overall goal.

3. **Product Increment:**
   A Product Increment is the usable and completed outcome of the work produced during a sprint.

It includes the newly completed Product Backlog items from the current sprint along with the increments of all previous sprints. Each new increment must work together with the previous increments and meet the team's Definition of Done. An increment is created whenever a Product Backlog item meets the Definition of Done. It does not necessarily have to be released to customers immediately. Product increments are incredibly useful in continuous integration and continuous delivery (CI/CD), as they allow teams to continuously integrate, validate, and deliver improvements to the product.

Commitment – Definition of Done:

The Definition of Done is a formal description of the quality standards and conditions that must be met before a product increment can be considered complete.It's important that teams have a clear and shared understanding of what "Done" means. For example, a development task may only be considered done when the code is implemented, tested, reviewed, and meets the required acceptance criteria.If a task does not meet the Definition of Done, it cannot be considered part of the completed increment and must be returned to the Product Backlog for future consideration.

Extended Artifacts:

In addition to the three official Scrum artifacts, there are some extended or supporting artifacts that teams use to monitor progress and maintain transparency.

1. **Burndown Chart:**
   Burndown charts are graphs that display the amount of remaining work over the duration of a sprint. Teams can examine in-progress burndown charts to identify whether they are on track to successfully complete the sprint.During the Sprint Review, teams can revisit the burndown chart to understand how their actual progress compared to their initial expectations.
2. **Definition of Done:**
   The Definition of Done is the shared understanding of the quality standards that a product increment must meet to be considered complete. It helps maintain consistency across the team and ensures that everyone understands when a task or feature is actually finished. When multiple Scrum teams work together on the same product, they should follow a common Definition of Done to maintain consistency and quality across the product.

Scrum Metrics:

Scrum metrics are specific data points that Scrum teams track and use to measure their progress, identify inefficiencies, and improve their overall effectiveness. They help teams understand how much work they can complete, whether they are meeting their sprint goals, and where improvements are needed in their development process.

Scrum metrics can be used to establish Key Performance Indicators (KPIs), but it depends on the type and scope of the work being measured.

While metrics help teams understand their performance, not every metric should be treated as a KPI. They should primarily be used to identify problems, improve processes, and support better decision-making rather than simply measuring individual performance.

There are many Commonly used metrics, some of which are:

1. **Sprint Burndown:**
   A Sprint Burndown Chart tracks the amount of remaining work throughout a sprint. It helps the team understand whether they are progressing at a reasonable pace to complete the sprint goal within the defined timeframe. If the chart shows that work is not being completed as expected, the team can investigate the reasons and adjust their plans accordingly.
2. **Team Velocity:**
   Velocity measures the amount of work a team completes during a sprint, usually measured in story points. It helps teams estimate how much work they can realistically take on in future sprints based on their previous performance. For example, if a team consistently completes around 30 story points per sprint, they can use that historical information to guide their future sprint planning. Velocity should primarily be used for planning within the same team rather than comparing the performance of different teams.
3. **Sprint Goal Success:**
   Sprint Goal Success measures whether the team achieved the main objective established during Sprint Planning. It helps determine whether the work completed during a sprint contributed to the intended outcome. It also allows teams to identify reasons behind unsuccessful sprint goals, such as unexpected blockers, poor estimation, or changing requirements.
4. **Cycle Time:**
   Cycle Time measures the amount of time taken to complete a task from the moment active work begins until it is finished. It helps teams understand how efficiently work moves through the development process. A shorter and more consistent cycle time can indicate that the team is completing tasks more efficiently, while unusually long cycle times may indicate bottlenecks or dependencies.
5. **Defect Rate:**
   Defect Rate measures the number of bugs or defects identified in a product during a particular period or development cycle. It helps teams understand the quality of their deliverables and identify areas that require improvement. Tracking defects also helps the team understand whether new features or changes are introducing unexpected problems.

Key Scrum Events and What Metrics to Review:

1. **Sprint Planning:**
   Sprint Planning is held at the beginning of a sprint to determine the work that the team can realistically complete.
   Some important metrics and data points to review include:
   1. Previous sprint velocity to understand how much work the team can handle.
   2. Team capacity to understand the availability of team members.
   3. Previous sprint completion rate to identify whether the team consistently meets its commitments.

    4. Sprint goals to ensure that the selected work contributes to the intended outcome.
2. **Daily Scrum:**
   The Daily Scrum is a short meeting where Developers inspect their progress towards the Sprint Goal and adjust their plans accordingly.
   Some important things to review include:
    1. Remaining work to understand whether the team is progressing as expected.
    2. Blockers and impediments that may prevent the team from completing its tasks.
    3. Tasks that are taking longer than expected.
    4. Dependencies that may affect the progress of other team members.
3. **Sprint Review:**
   During the Sprint Review, the Scrum Team and stakeholders inspect the completed increment and discuss progress towards the product goal.
   Some important metrics to review include:
    1. Completed Product Backlog items to understand what was delivered.
    2. Sprint Goal achievement to evaluate whether the intended outcome was reached.
    3. Defect rate to understand the quality of the delivered increment.
    4. Stakeholder feedback to identify further improvements and changes in priorities.
4. **Sprint Retrospective:**
   During the Sprint Retrospective, the Scrum Team reflects on the previous sprint and identifies ways to improve its working process.
   Some important metrics to review include:
    1. Sprint completion rate to understand how much planned work was completed.
    2. Cycle time to identify delays and bottlenecks.
    3. Number of blockers and the time taken to resolve them.
    4. Defect trends to identify recurring quality issues.
5. The purpose of reviewing these metrics is to identify areas of improvement and create actionable plans for the upcoming sprints.

Scrum of Scrums:

Scrum of Scrums is a scaled Agile technique that offers a way to connect multiple Scrum teams who need to work together to deliver complex solutions.It is a coordination technique used when multiple Scrum teams work together on a larger project, enabling cross-team collaboration, communication, and alignment.

While a regular Daily Scrum focuses on the progress and blockers within a single Scrum team, Scrum of Scrums focuses on coordinating the work of multiple teams and identifying dependencies that may affect the overall project.

The Purpose of Scrum of Scrums:

When multiple Scrum teams work on different parts of the same product, their work may depend on one another. Without proper coordination, delays or blockers within one team can affect the progress of other teams. Scrum of Scrums helps address these challenges by bringing representatives from different teams together to discuss their progress, upcoming work, dependencies, and impediments.

It helps ensure that:

1. Different teams remain aligned towards a common product goal.
2. Cross-team dependencies are identified and managed early.
3. Blockers affecting multiple teams are communicated and resolved.
4. Teams maintain transparency regarding their progress and responsibilities.
5. The overall project moves forward without unnecessary delays.

How Scrum of Scrums Works:

Scrum of Scrums usually takes place after the individual teams have completed their Daily Scrum meetings. Each Scrum team selects a representative, often a Scrum Master, technical lead, or another team member who understands the team's progress and dependencies. These representatives participate in a scaled Daily Scrum, where they discuss the overall progress of the teams and identify issues that require cross-team collaboration. The representatives communicate the relevant information back to their respective teams so that everyone remains aligned.

The meeting generally focuses on three questions:

1. What has our team completed since the last Scrum of Scrums meeting?
2. What will our team work on before the next Scrum of Scrums meeting?
3. Are there any blockers, risks, or dependencies that may affect other teams?

When teams are ready, here are some considerations that may be useful:

1. Keep the scaled Daily Scrum meeting to 15 minutes, mirroring the individual team Daily Scrum.
2. Conduct the scaled Daily Scrum shortly after the last individual team Daily Scrum.
3. Establish a working agreement for the Scrum of Scrums.
4. Agree on the collective and individual Definition of Done and make sure that all teams share and follow it.
5. Establish a routine or agenda to keep the scaled Daily Scrum focused.
6. Start tracking the number of days teams remain blocked by impediments.
7. Track how many scaled Daily Scrums were started and finished on time.
8. Focus on delivering stories that have dependencies first to reduce risk and enable other teams to continue their work.
9. Track and visualize the number of days remaining until the next demo meeting.

The overall purpose of Scrum of Scrums is to improve communication and coordination between multiple Scrum teams, ensuring that they can work together effectively towards delivering a common product.

Resources:

1. **https://www.atlassian.com/agile**
2. **https://www.atlassian.com/agile/scrum**
3. **https://www.geeksforgeeks.org/software-testing/what-is-agile-methodology/**

4. https://www.geeksforgeeks.org/software-engineering/scrum-software-development/
5. https://www.scrum.org/resources/what-is-a-product-owner
6. https://www.scrum.org/resources/what-is-a-scrum-master
7. https://www.scrum.org/resources/what-is-a-sprint-retrospective
8. https://resources.scrumalliance.org/Article/scrum-artifacts
9. https://www.scrumatscale.com/scrum-at-scale-guide-online/
10.